

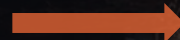
THE INTELLIGENCE LAYER FOR MARS

Harnessed agentic AI for science, engineering and settlement

SCIENCE



ENGINEERING



OPERATIONS

OCTOBER 22–24, 2026 • USC, LOS ANGELES

BOB J MILLS



Mars is a learning-rate problem

The scarce resource is elapsed time from uncertain observation to a verified decision.



21–23 min

maximum one-way
communication delay in
NASA crewed-Mars
profiles

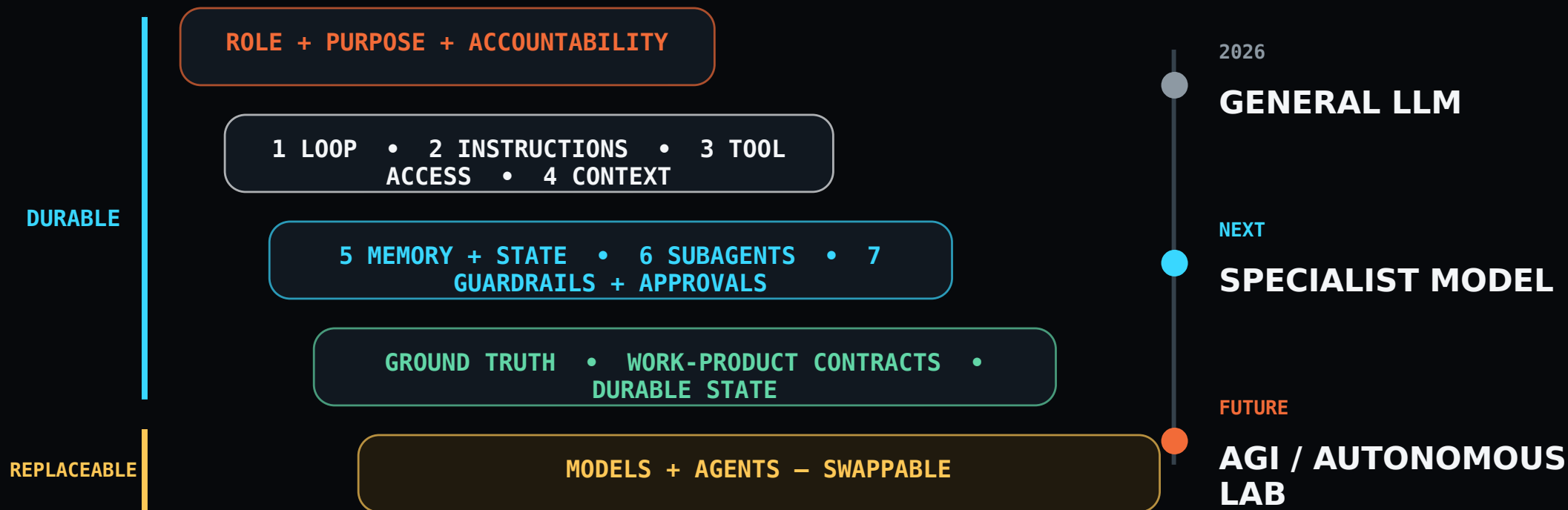
13–21 days

blackout examples in
two representative NASA
mission profiles

A harness attacks coordination latency while preserving the evidence needed to close the decision.

Seven functions turn a model into a mission harness

The model supplies reasoning. Seven surrounding functions make its work executable, reviewable and durable.



Stable contracts and approval authority let every stronger model inherit a mission-ready way to work.

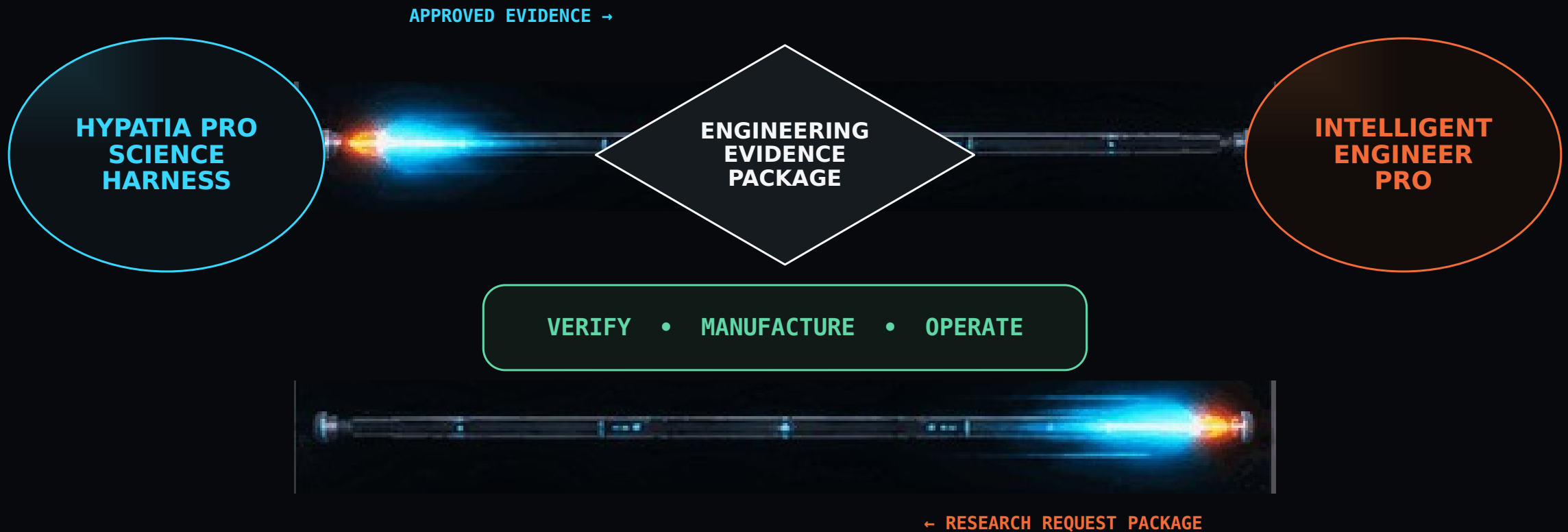
Skills are durable; agents are assignments

Define the capability, evidence and work product first. Then assign the best executor available.



Two harnesses. One digital thread.

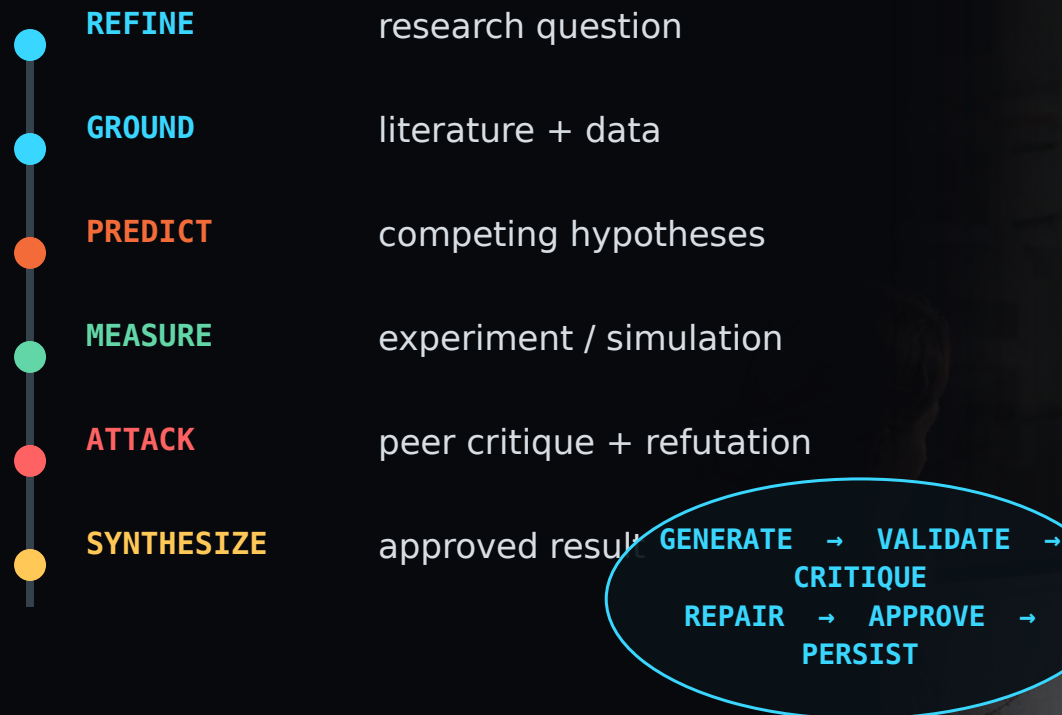
Research and engineering stay specialized, but evidence and unresolved questions move through one governed baseline.



**Field and test evidence can reopen the science
—or the design.**

Hypatia turns curiosity into defensible science

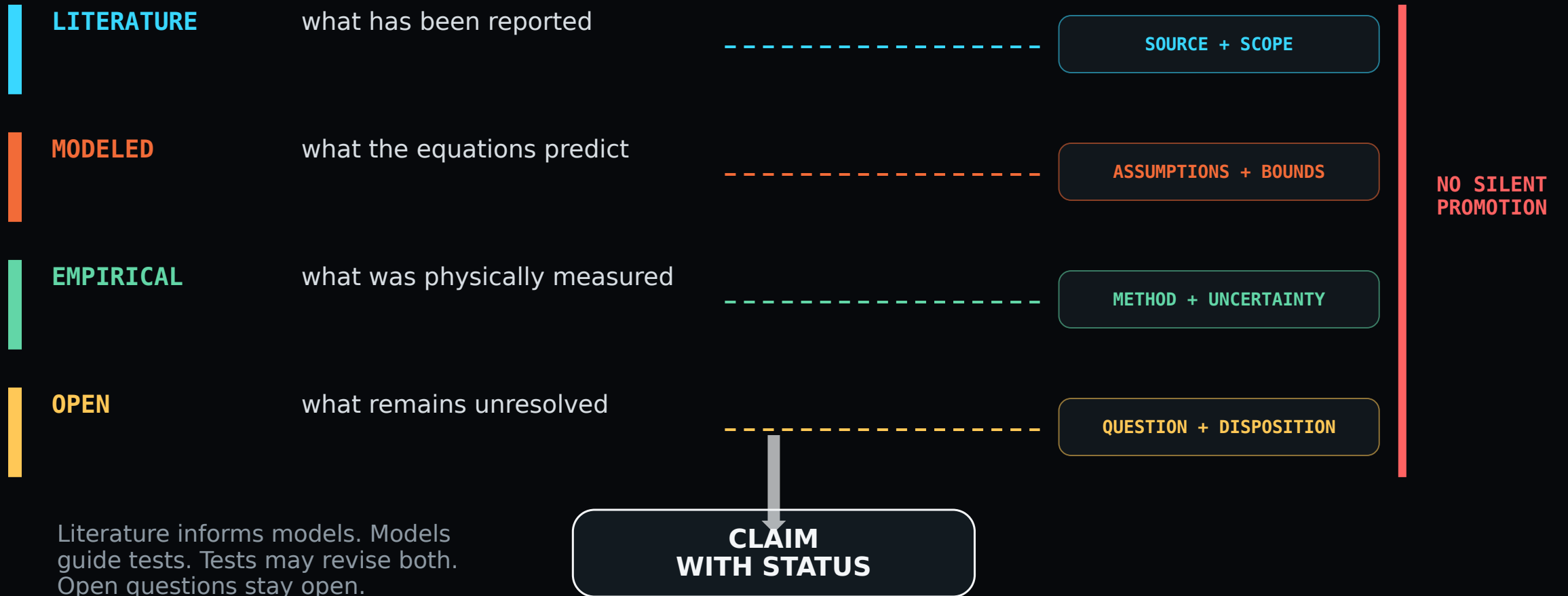
Falsifiable questions. Explicit variables. Classified evidence. Adversarial attack.



ONLY APPROVED OUTPUTS FLOW FORWARD

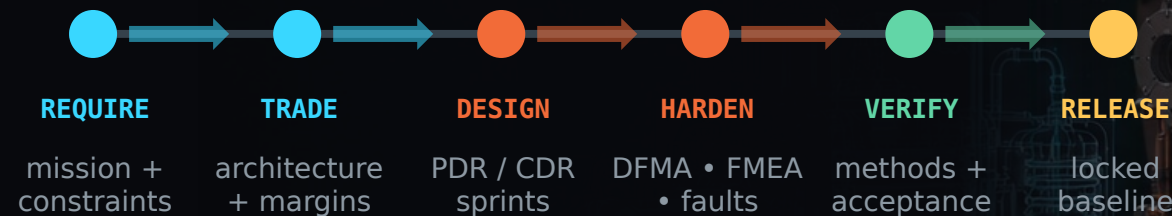
Every claim carries epistemic status

The harness prevents a plausible sentence—or a clean simulation—from silently becoming a physical fact.



Intelligent Engineer turns evidence into a controlled baseline

Drafts accelerate work. Phase gates authorize work.

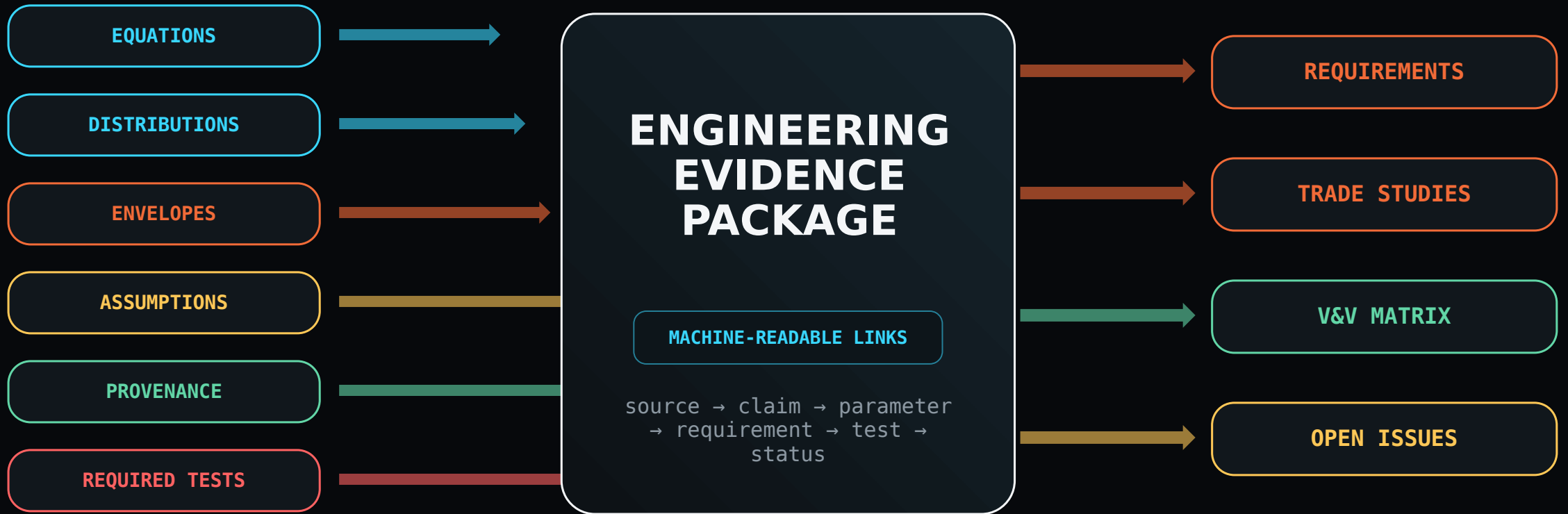


LOCKED BASELINE

Only reviewed phase outputs become authorized inputs downstream.

The handoff is structured—not conversational

Engineering inherits equations, bounds, provenance and tests together, so changed evidence can reopen the right decisions.



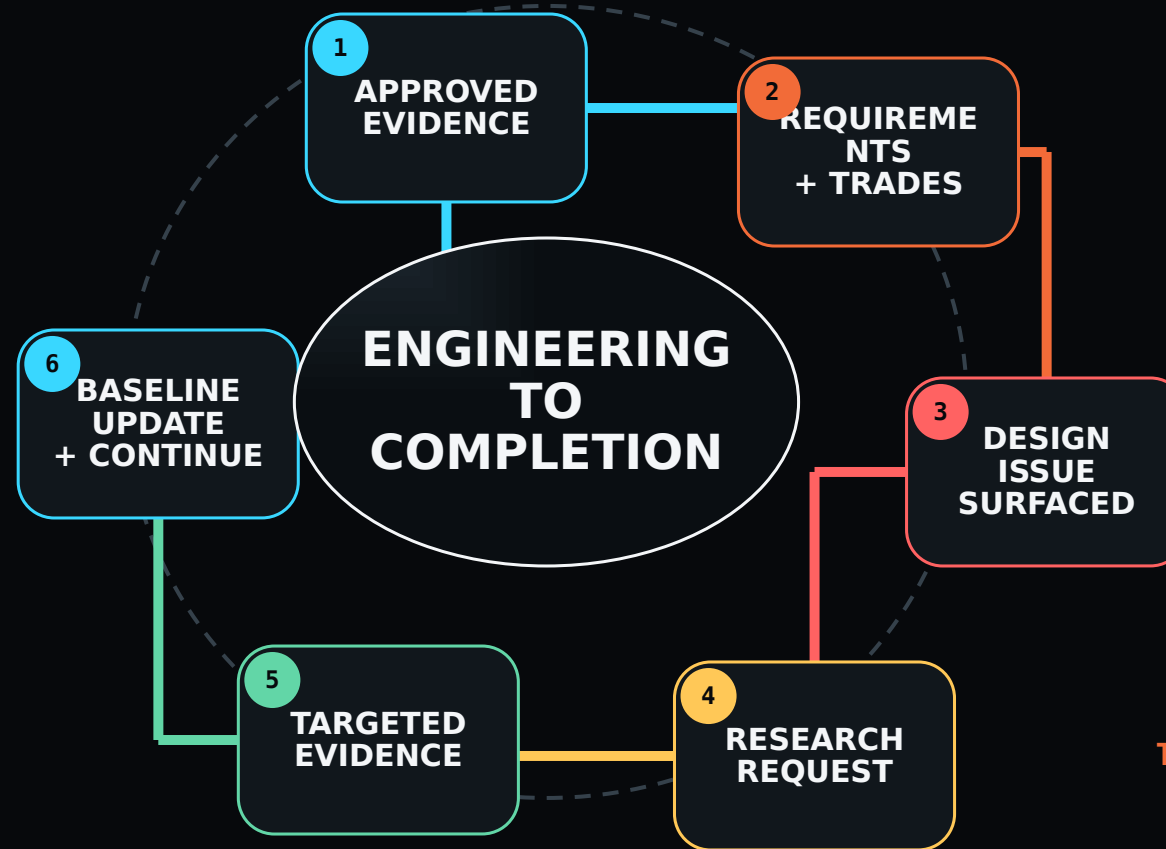
**A changed parameter triggers
focused rework—not a search through
old chat.**

The fastest path to completion includes a return to research

Agents accelerate the issue → research request → evidence return → baseline update cycle instead of hiding uncertainty.

NO FAKE CERTAINTY

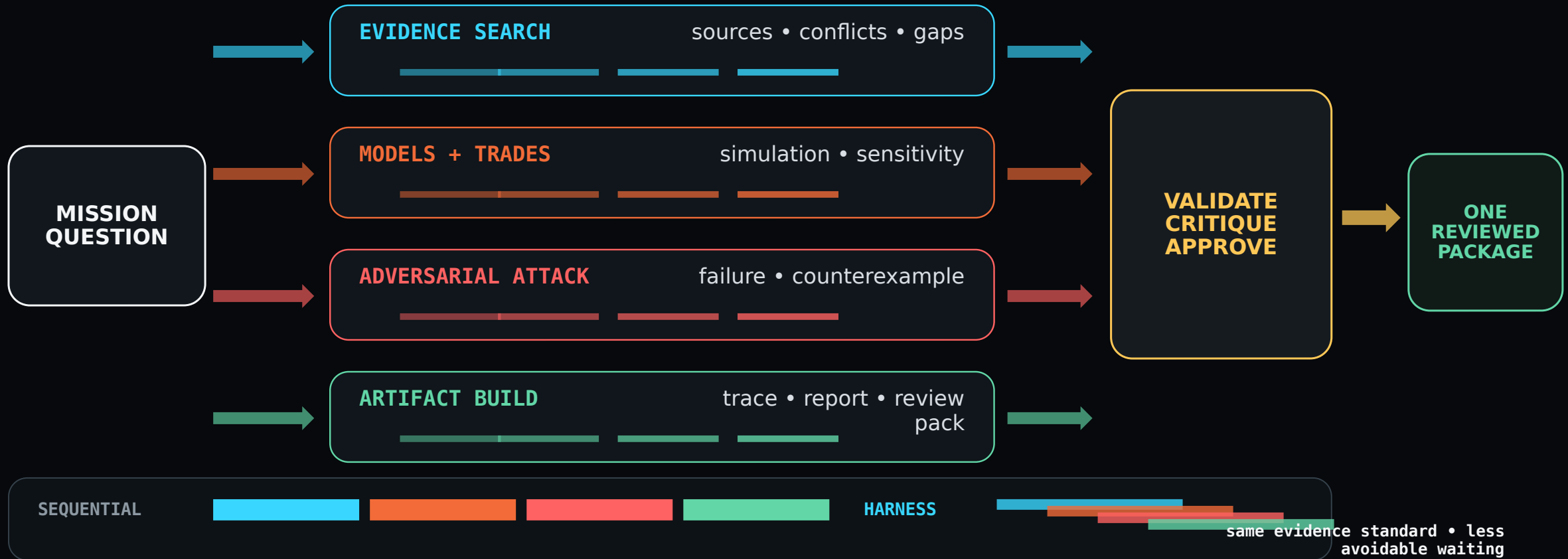
Verification,
manufacturing and
operations can reopen
any earlier node.



THE RETURN PATH IS A FEATURE

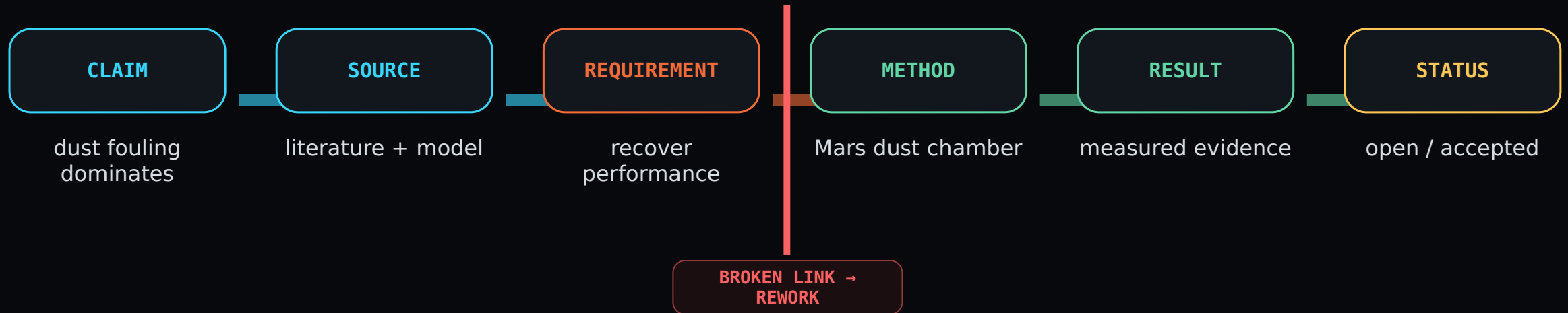
Parallel work shortens waiting; gates preserve truth

The harness coordinates independent searches, analyses and critiques—then merges only reviewed work.



Verification starts at the first claim

Agents maintain the links, expose breaks and launch focused rework before a review gate.



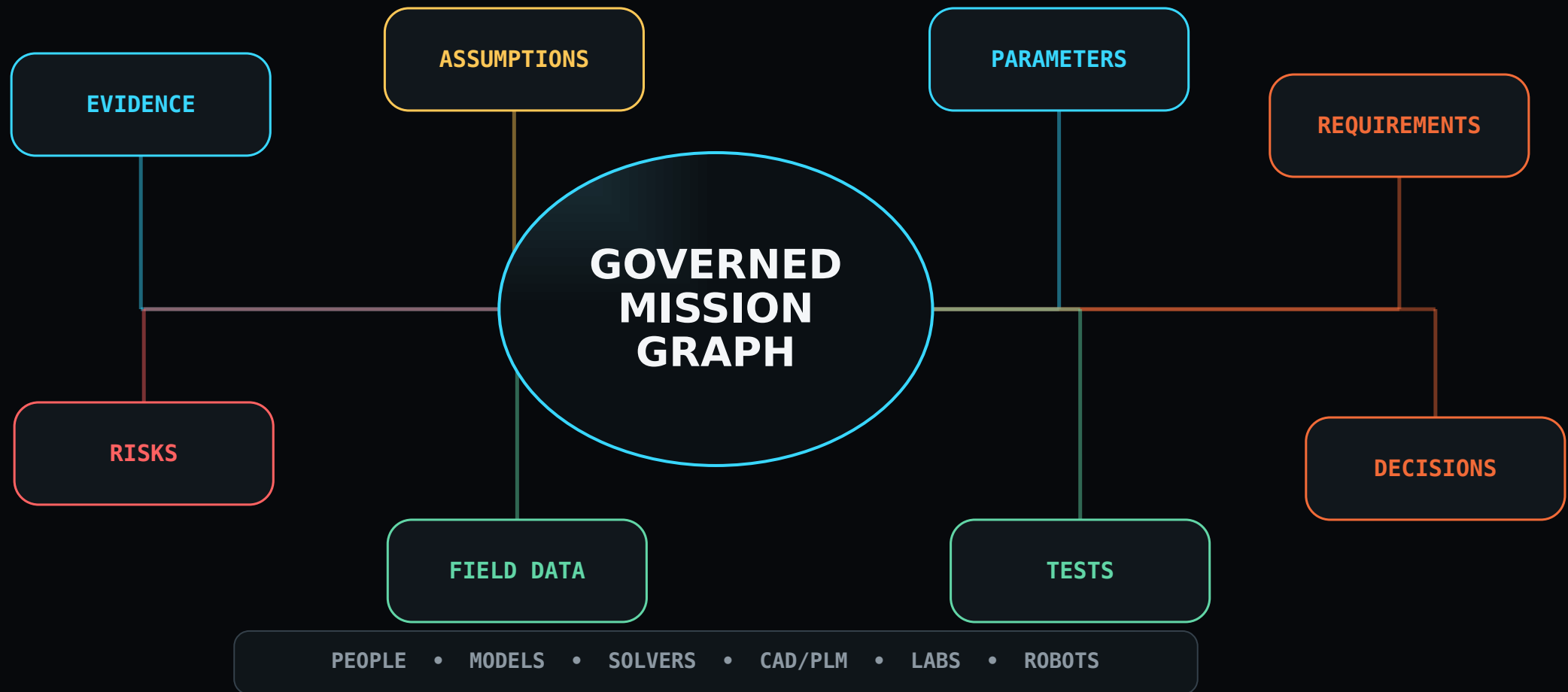
Every uncertainty gets a disposition



AI can assist verification. It cannot make missing physical evidence real.

Mission memory is the compounding asset

A governed knowledge graph preserves what the mission knows, assumes, decides, tests and learns across model generations.



Mars cannot wait for Earth to think

Delay and blackout make local intelligence mission infrastructure—not a convenience.

21–23 min

maximum one-way delay
in NASA crewed-Mars
profiles

WEEKS

communications
blackouts are
unavoidable in
representative profiles

LOCAL

operate through
delay

SYNC

reconcile with Earth

SAFE

degrade predictably

Cached evidence • controlled tools • asynchronous
synchronization • explicit authority

MOXIE proved the chemistry. Settlement scale is a systems problem.

EMPIRICAL — NASA MOXIE

16

oxygen
production
runs

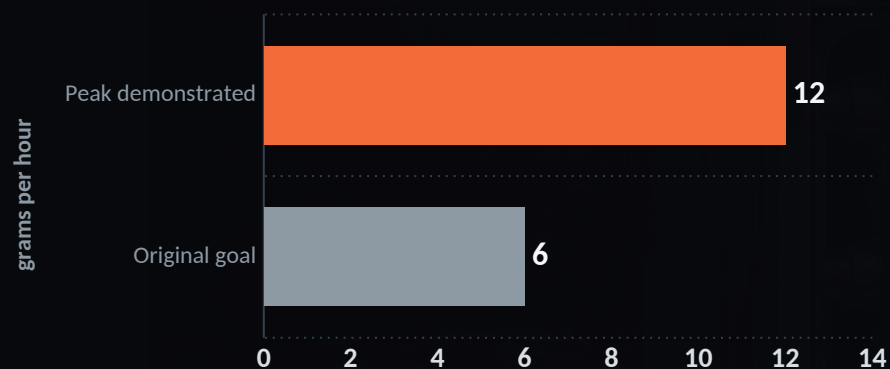
Peak O₂ production rate

122 g

total oxygen
produced

≥98%

peak oxygen
purity



OPEN FOR SETTLEMENT SCALE

dust • inlet variability • thermal
cycling • aging • liquefaction •
storage • maintenance

Evidence becomes a requirement—or remains an assumption

Illustrative translation of the ISRU evidence package. This is not a qualified design baseline.

ILLUSTRATIVE • REPLACE WITH REVIEWED HARNESS OUTPUT

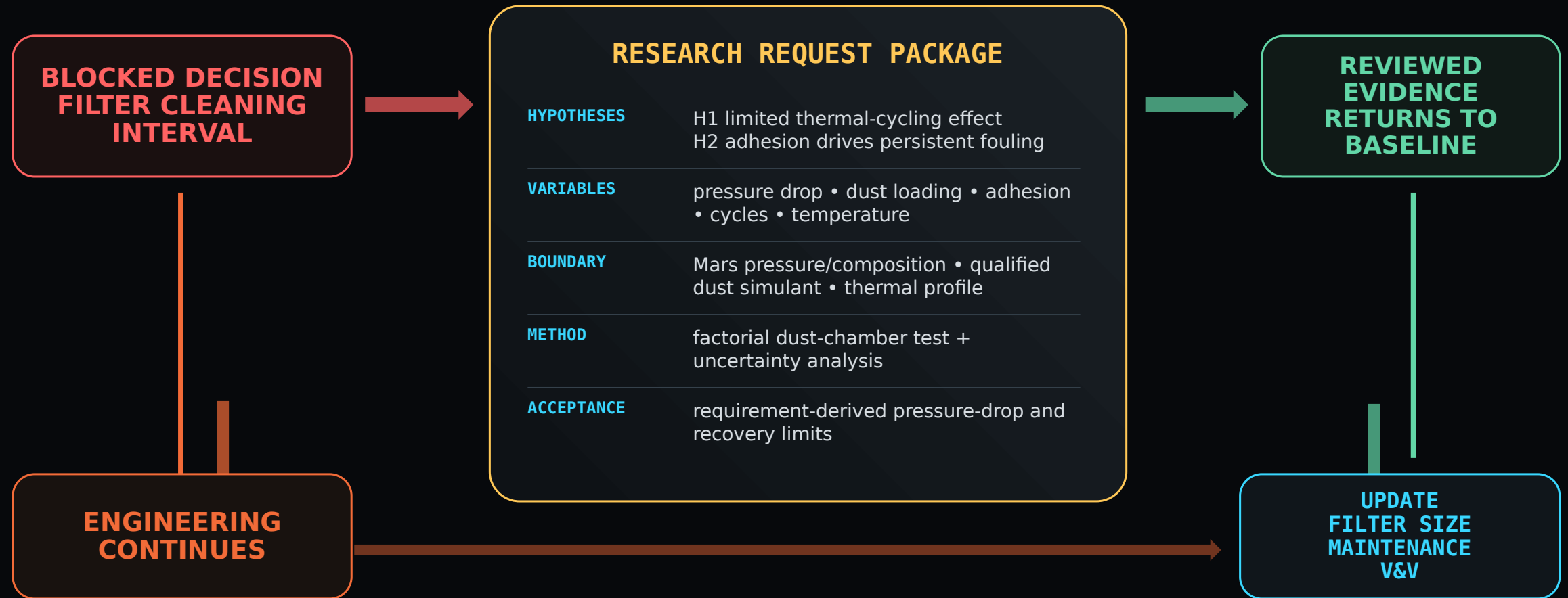


Every row keeps the source, obligation and proof method connected.

Engineering returns one bounded scientific question

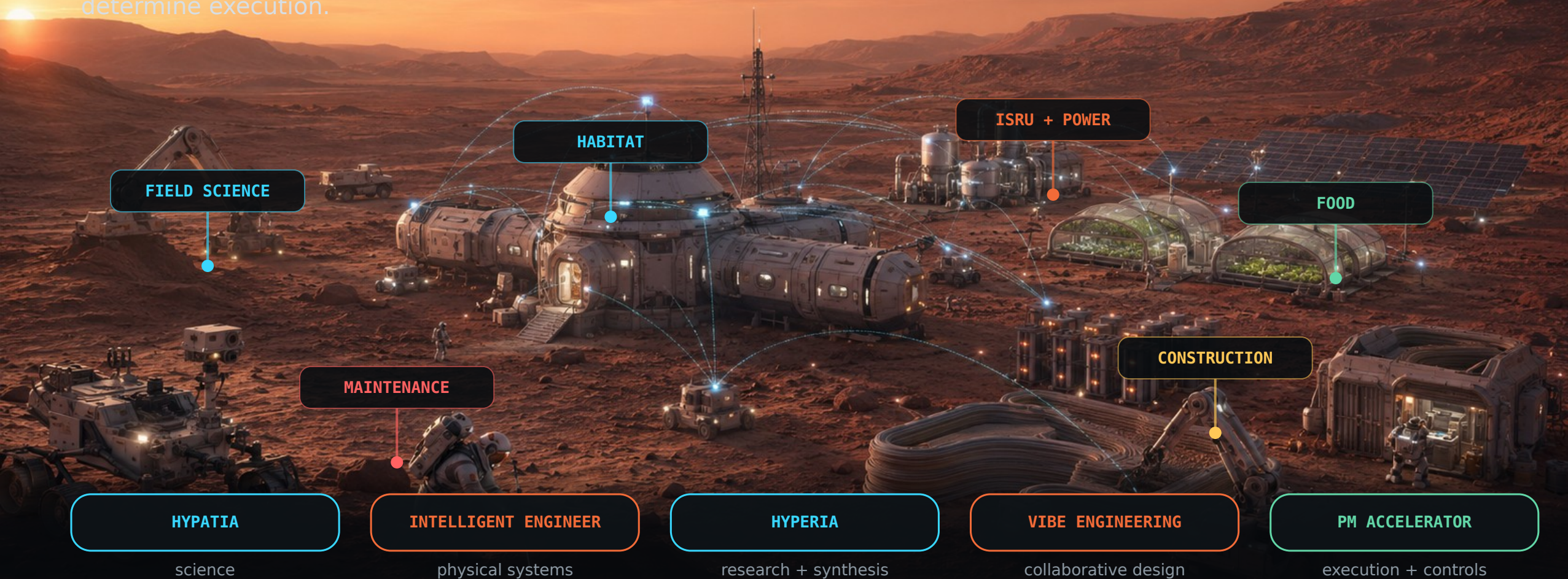
A design-blocking uncertainty becomes an executable Research Request Package—not a vague demand for more study.

ILLUSTRATIVE RESEARCH REQUEST • NOT AN EMPIRICAL RESULT



SaaS workflows become portable mission skills

Codex, Claude Cowork, Hermes and Grok can load the same core skill contracts; available tools determine execution.



RUNTIMES: CODEX • CLAUDE COWORK • HERMES • GROK

MAKE ENGINEERING A LEARNING SYSTEM

Mars will reward teams that learn faster without losing discipline.



Build the intelligence layer now—so every stronger model inherits a mission-ready way to work.